

EventKD: Event-Aware Knowledge Distillation for Generative Event Extraction

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Abstract

Event extraction aims to identify triggers in text, classify them into predefined event types, and recognize associated arguments such as participants and locations. Large language model (LLM)-based generative approaches provide a unified paradigm for joint trigger identification, classification, and argument extraction; however, their computational cost necessitates knowledge distillation (KD) into smaller student models. Existing KD methods for LLM distillation operate at the token level, treating each output position independently and thus failing to concentrate the distillation signal on event-relevant spans or to capture the relational structure among event components. We propose **EventKD**, a two-level distillation framework tailored for event extraction. At the token level, a distributional divergence loss incorporates event-aware span importance weighting, derived from the teacher’s structured output, to prioritize tokens corresponding to triggers, arguments, event type information. At the span level, an Event-Aware (EA) loss aligns pairwise distance structures among event-aware span representations between teacher and student across multiple Transformer layers. Experiments on three benchmarks (ACE05, GENEVA and MAVEN) demonstrate that EventKD consistently outperforms existing distillation methods, e.g., improving trigger F1 by 2.25 points and argument F1 by 5.32 points on ACE05 over the strongest baseline.

1 Introduction

Event Extraction (EE) is a core task in information extraction concerned with identifying event triggers in text, classifying them into predefined event types, and recognizing the arguments associated with each event, such as participants, locations, and temporal attributes (Doddington et al., 2004). Accurate event extraction underpins downstream applications ranging from knowledge graph construction and intelligence analysis to biomedical

literature mining, making it a longstanding focus of NLP research (Wang et al., 2020; Li et al., 2021).

Recent advances have recast event extraction as a generative task in which a language model produces structured outputs conditioned on the input text, offering a unified paradigm for joint trigger identification, classification, and argument extraction (Du and Cardie, 2020; Lu et al., 2021; Li et al., 2021). Although large language models (LLMs) achieve strong performance within this paradigm, their substantial computational cost renders direct deployment impractical in many real-world settings. Knowledge distillation (KD) provides a principled solution: a frozen LLM teacher guides the training of a compact student model, transferring the teacher’s rich event representations into an efficient architecture suitable for deployment.

Existing KD methods for LLMs, however, are not tailored to the demands of event extraction. Standard KL divergence minimization (Hinton et al., 2015) is susceptible to mode averaging and gradient instability (Wen et al., 2023). Recent methods address these shortcomings through alternative divergences (Wen et al., 2023; Ko et al., 2024; Kim et al., 2026), on-policy and contrastive training (Agarwal et al., 2024; Ko et al., 2025), and structural relationship transfer (Park et al., 2019). Despite these advances, all such methods operate at the token level and treat each output position independently, failing to (1) concentrate the distillation signal on event-relevant tokens such as triggers and arguments, and (2) capture the relational structure among event components that is essential for accurate event understanding.

To address these limitations, we propose **EventKD**, a two-level knowledge distillation framework designed specifically for event extraction. At the token level, we employ a general distributional divergence loss $\mathcal{L}_{\text{KD}}$ with event-aware span importance weighting, which concentrates the distillation signal on tokens corresponding to triggers, arguments,

and event types. At the span level, we introduce an Event-Aware (EA) Knowledge Distillation loss that preserves pairwise geometric relationships among event-aware spans across multiple Transformer layers. By combining both objectives, our framework enables a compact student to capture not only the teacher’s output behavior but also the internal structural knowledge critical for event understanding. Our contributions are as follows:

- We propose EventKD, a knowledge distillation framework that transfers structured event knowledge from a large teacher LLM to a compact student model, preserving both token-level output behavior and span-level relational structure.
- We introduce an event-aware knowledge distillation approach comprising two complementary components: (1) a general token-level distillation loss $\mathcal{L}_{\text{KD}}$ with span importance weighting that focuses the distillation signal on event-relevant positions, and (2) an Event-Aware distillation loss that preserves pairwise structural relationships among event-aware spans across network layers. The framework is agnostic to the choice of divergence; in our experiments, we instantiate $\mathcal{L}_{\text{KD}}$ with the Skew Forward KL (SFKL) divergence.
- We conduct extensive experiments on three benchmarks (ACE05, GENEVA and MAVEN), demonstrating consistent improvements over existing distillation methods.

2 Related Work

LLM-based Event Extraction. Recent advances have shifted event extraction from discriminative classification to generative formulations (Du and Cardie, 2020; Lu et al., 2021; Li et al., 2021), a trend amplified by LLMs that leverage in-context learning and instruction following (Srivastava et al., 2025; Liu et al., 2025). Despite their strong performance, LLM-based event extractors suffer from critical limitations: they operate purely at the token level without modeling structural relationships between event components, and their computational cost prohibits deployment in resource-constrained settings. While recent reasoning-enhanced approaches (Srivastava et al., 2025; Liu et al., 2025) improve over baseline LLMs through divergent-convergent reasoning chains, they remain confined

to a single model’s output distribution and do not transfer structural event knowledge to efficient architectures.

Knowledge Distillation. Conventional KD approaches focus primarily on minimizing output-level divergence while overlooking structural knowledge in intermediate representations (Hinton et al., 2015). Feature-based methods address this gap by aligning internal hidden states or attention maps between student and teacher (Jiao et al., 2019). Recent LLM distillation research has further explored on-policy training (Agarwal et al., 2024), contrastive objectives (Ko et al., 2025), dual-space alignment for cross-tokenizer settings (Zhang et al., 2024), efficient sparse logit methods (Anshumann et al., 2025), and discrete score-matching objectives that bypass softmax smoothing (Kim et al., 2026); see Xu et al. (2024) for a comprehensive survey. Nevertheless, most existing approaches adopt a static, uniform alignment strategy at the token level, neglecting which tokens carry event-specific semantics and how event components relate structurally.

Our work addresses these gaps by distilling structural, span-level event knowledge from a frozen LLM teacher into a compact student model. Crucially, the distillation operates at the *span level*, explicitly aligning the pairwise relational geometry of event-aware spans (triggers, arguments, and types) between teacher and student. This **span-aware, event-aware, relation-aware** distillation preserves the structural topology of event representations that purely generative approaches discard, while simultaneously reducing inference cost.

3 Preliminaries

3.1 Problem Formulation and LLM-based Event Extraction

Problem Formulation. Given an input sentence $\mathbf{x} = (x_1, x_2, \dots, x_N)$ of N tokens, the goal of event extraction is to identify event triggers, classify them into predefined event types from $\mathcal{E}$, and extract their associated arguments. Each extracted event is represented as $e_k = (t_k, c_k, \{(a_{k,j}, r_{k,j})\}_{j=1}^{J_k})$, where t_k is the event trigger (the word or phrase that most clearly evokes the event), $c_k \in \mathcal{E}$ is the event type, and $\{(a_{k,j}, r_{k,j})\}$ denotes the associated arguments with $a_{k,j}$ being an argument span and $r_{k,j} \in \mathcal{R}$ its semantic role (e.g., Agent, Place, Time) (Ebner et al., 2020). Given the full set of extracted events

$\mathcal{Y} = \{e_1, e_2, \dots, e_M\}$ (with $\mathcal{Y} = \emptyset$ when no event is present), the structured event representation naturally motivates a span-level approach to knowledge distillation, as we describe in §4.1.

LLM-based Event Extraction. We employ a language model as the Teacher, which performs event extraction using structured prompts. The Teacher is fine-tuned on each dataset using parameter-efficient adaptation (see §5.1 for details). Specifically, given an input sentence $\mathbf{x}$, we construct a prompt $P(\mathbf{x})$ that includes a task instruction describing the event extraction schema, the set of target event types $\mathcal{E}$ with their associated argument roles $\mathcal{R}$, and the input sentence. The Teacher LLM autoregressively generates a structured output sequence $\mathbf{y} = (y_1, y_2, \dots, y_T)$ that encodes the extracted events in a linearized format:

$$\mathbf{y} = f_{\text{LLM}}(P(\mathbf{x}); \theta_T), \quad (1)$$

where θ_T denotes the Teacher parameters and $\mathbf{y}$ is decoded into the structured event set $\mathcal{Y}$. The output sequence $\mathbf{y}$ naturally contains identifiable spans corresponding to event components: trigger spans, argument spans, and event type regions that encompass complete event annotations. Figure 1 illustrates the prompt design. These spans serve as the foundation for our Event-Aware distillation approach described in §4.1. The model is trained by minimizing the standard autoregressive cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{t=1}^T \log p_{\theta}(y_t | \mathbf{y}_{<t}, P(\mathbf{x})), \quad (2)$$

where θ denotes the model parameters and $p_{\theta}(y_t | \mathbf{y}_{<t}, P(\mathbf{x}))$ is the predicted probability of the ground-truth token y_t given the preceding tokens and the prompt. This objective applies to both teacher fine-tuning ($\theta = \theta_T$) and student training ($\theta = \theta_S$).

3.2 Knowledge Distillation

Knowledge Distillation (KD) (Hinton et al., 2015) transfers knowledge from a large teacher model to a compact student model by minimizing the divergence between their output distributions. For autoregressive language models, the sequence-level divergence decomposes as a sum over token-level conditionals. Specifically, for a training dataset $\mathcal{D} = \{(\mathbf{x}, \mathbf{y})\}$, the standard forward KL divergence

loss is:

$$D_{\text{KL}}(p_T \| p_S) \approx \frac{1}{|\mathcal{D}|} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}} \sum_{t=1}^{|\mathbf{y}|} \sum_{v \in \mathcal{V}} p_{T,t}^v \log \frac{p_{T,t}^v}{p_{S,t}^v}, \quad (3)$$

where $p_{T,t}^v = p_T(v | \mathbf{y}_{<t}, \mathbf{x})$ and $p_{S,t}^v = p_S(v | \mathbf{y}_{<t}, \mathbf{x})$ denote the teacher and student conditional distributions over the vocabulary $\mathcal{V}$ at time step t , respectively. This formulation enables the student to approximate the teacher’s distribution and capture inter-class correlations (Hinton et al., 2015).

The choice of divergence function significantly affects distillation quality. Forward KL encourages the student to cover all modes of the teacher distribution but suffers from gradient instability when the student assigns near-zero probability to teacher-favored tokens. Reverse KL (RKL) (Wen et al., 2023) instead minimizes $D_{\text{KL}}(p_S \| p_T)$, producing mode-seeking behavior that yields sharper student distributions but risks ignoring parts of the teacher’s support. Skew Forward KL (SFKL) (Ko et al., 2024) interpolates a mixture distribution $p_M = \lambda \cdot p_T + (1 - \lambda) \cdot p_S$ into the forward KL target, yielding provably more stable gradients while retaining the mode-covering property. More recently, Concrete Score Distillation (CSD) (Kim et al., 2026) reformulates the distillation objective through a score-matching framework over continuous relaxations of discrete tokens, bypassing the pitfalls of direct distribution matching. We evaluate all three alternatives alongside standard forward KL in our experiments (§5).

4 Event-Aware Knowledge Distillation

This section presents our knowledge distillation framework for LLM-based event extraction. Building on the task formulation and Teacher LLM described in §3.1, we introduce Event-Aware Knowledge Distillation, which transfers structured event knowledge from teacher to student through span-level relational alignment (§4.1), and present the overall training objective (§4.2). An overview of the full framework is provided in Figure 3 (Appendix).

4.1 Event-Aware Knowledge Distillation Loss

As illustrated in Figure 2, in LLM-based event extraction (Liu et al., 2025; Srivastava et al., 2025), the Teacher LLM implicitly performs *event identification* (localizing trigger and argument spans within the input) and *event classification* (assigning each trigger to an event type from the schema

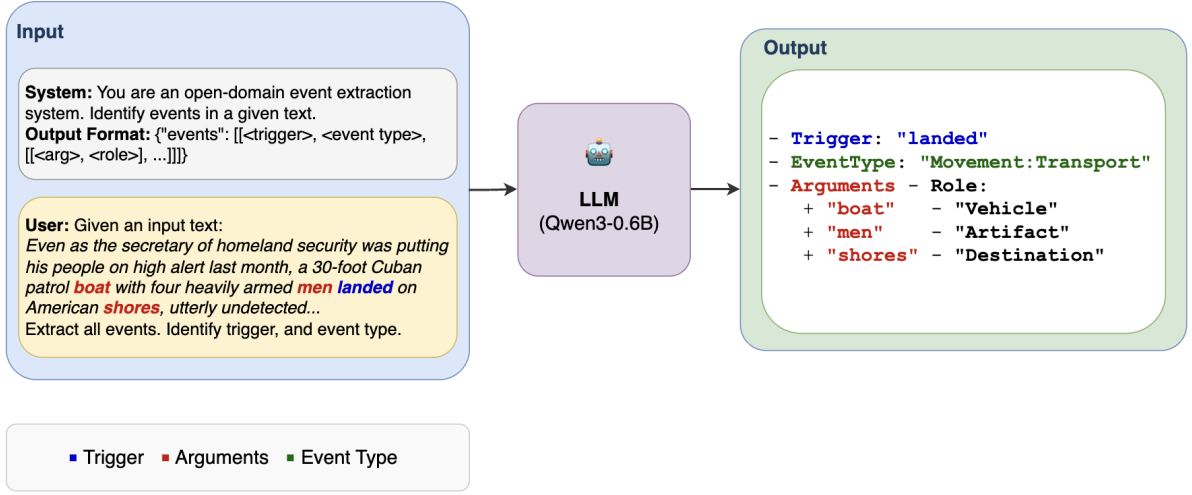


Figure 1: Prompt design for standard event extraction. The model receives the input text and must identify all event triggers and classify their event types.

$\mathcal{E}$). These two operations leave distinct footprints in the model’s hidden states: identification concentrates attention on salient token spans, while classification encodes type-discriminative features in higher-layer representations.

Our Event-Aware Knowledge Distillation framework is designed to capture both signals through two complementary objectives. At the *token level*, a distributional divergence loss $\mathcal{L}_{\text{KD}}$ transfers the teacher’s output distribution, preserving the identification and classification decisions encoded in the predicted token probabilities. At the *span level*, our Event-Aware loss $\mathcal{L}_{\text{EA}}$ preserves the geometric structure among event-aware spans, ensuring that the student internalizes not just individual span representations but the full relational web that characterizes an event instance.

Motivation. Conventional generative knowledge distillation forces the student to mimic the teacher’s output token by token in a left-to-right autoregressive manner. While this captures surface-level distributional knowledge, it overlooks the rich structural information inherent in event extraction. Events possess an intrinsic graph-like structure: each trigger is associated with a specific event type, and arguments depend on both the trigger and their semantic roles. These relational dependencies form a complex web of associations that a purely sequential distillation objective cannot adequately preserve. We therefore propose an *Event-Aware Knowledge Distillation* framework that explicitly transfers the structural event knowledge from the teacher to the student through span-level relational

alignment.

Span Representation. To capture the structured nature of events, we define three categories of event-aware spans within the generated output sequence: (1) *Trigger Spans*, the token positions corresponding to event triggers that express event occurrences; (2) *Argument Spans*, the token positions of entities or temporal expressions filling semantic roles in an event (Ebner et al., 2020); and (3) *Event Type Spans*, the token positions of verbalized event type labels in the output.

For each span k at layer l , we compute its representation $U_{k,l}$ as a weighted average of constituent token hidden states:

$$U_{k,l} = \frac{\sum_{t \in \mathcal{S}_k} w_{t,l} \cdot H_{t,l}}{\sum_{t \in \mathcal{S}_k} w_{t,l}}, \quad (4)$$

where $\mathcal{S}_k$ is the set of token indices in span k , $H_{t,l}$ is the hidden state of token t at layer l , and $w_{t,l}$ is a token importance weight. We derive $w_{t,l}$ from a pairwise relevance matrix A_l computed from hidden states by measuring the weighted in-degree of each token (excluding self-loops), followed by standardization across the sequence:

$$w_{t,l} = \frac{\tilde{w}_{t,l}}{\sum_{t'=1}^N \tilde{w}_{t',l}}, \quad \tilde{w}_{t,l} = \sum_{s \neq t} \alpha_{s \rightarrow t,l}, \quad (5)$$

where $\alpha_{s \rightarrow t,l}$ is the pairwise relevance score from token s to token t at layer l (see Appendix A.1 for derivation). Tokens that receive higher attention from other positions are deemed more semantically important within their span.

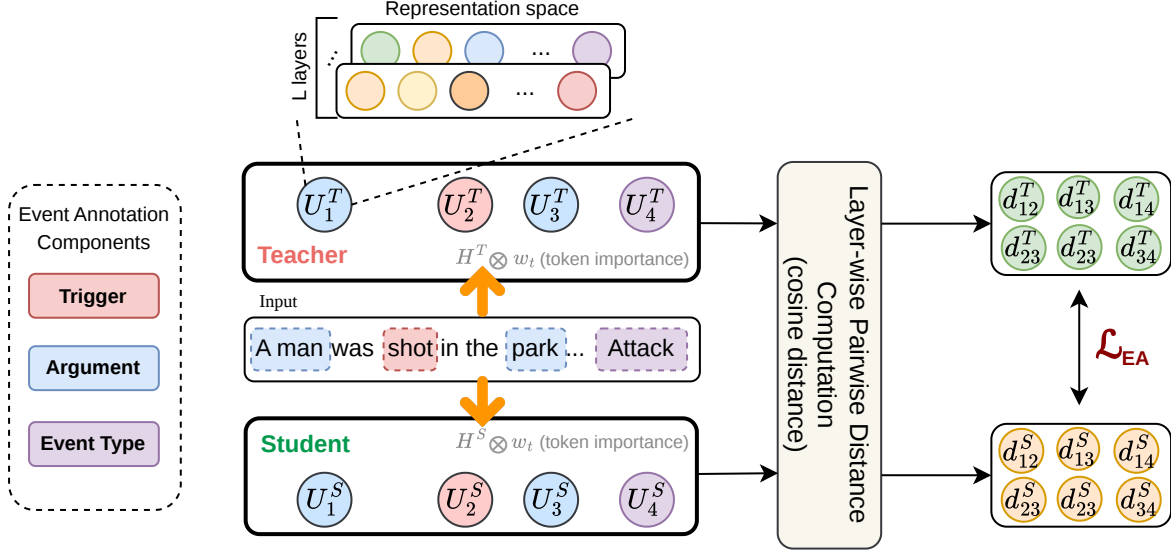


Figure 2: Illustration of Event-Aware Knowledge Distillation. Event-aware spans (triggers, arguments, and event types) are extracted from both teacher and student hidden states. Pairwise cosine distances d_{ij} between spans form a relational structure within each model. The Event-Aware loss $\mathcal{L}_{EA}$ aligns these pairwise distances, preserving the structural geometry of event representations across teacher and student.

Event-Aware Loss. The core of our approach is an event-aware loss that preserves the pairwise geometric structure among event spans across teacher and student. Rather than matching individual span representations directly, we align the *relational geometry*, that is, how triggers relate to arguments and how different event types are positioned relative to one another in representation space. This relational formulation is inherently more robust to the capacity gap between teacher and student, as it requires the student to reproduce structural relationships rather than absolute representations.

We compute span-level importance weights by aggregating token weights:

$$w_{i,l}^{\text{sp}} = \frac{\tilde{w}_{i,l}^{\text{sp}}}{\sum_{j=1}^{N_l^{\text{sp}}} \tilde{w}_{j,l}^{\text{sp}}}, \quad \tilde{w}_{i,l}^{\text{sp}} = \sum_{t=s_i^l}^{e_i^l} w_{t,l}^{(T)}, \quad (6)$$

where $[s_i^l, e_i^l]$ denotes the token indices of span i , N_l^{sp} is the total number of event-aware spans at layer l , and $w_{t,l}^{(T)}$ is the teacher-derived token importance weight.

Following prior work on intermediate-layer distillation (Romero et al., 2015; Sun et al., 2019), we select a subset of Transformer layers $L_{\text{key}} \subset \{1, \dots, L\}$, where L is the total number of Transformer layers, from which to extract span representations (the specific layers are treated as a hyperparameter; see §5 for details). For these key layers,

the event-aware loss is defined as:

$$\mathcal{L}_{EA} = \frac{1}{|L_{\text{key}}|} \sum_{l \in L_{\text{key}}} \sum_{i=1}^{N_l^{\text{sp}}} \sum_{j=i+1}^{N_l^{\text{sp}}} w_{ij,\phi(l)}^{\text{sp}} \cdot \left| d(U_{i,\phi(l)}^T, U_{j,\phi(l)}^T) - d(U_{i,l}^S, U_{j,l}^S) \right|, \quad (7)$$

where $d(\cdot, \cdot)$ denotes cosine distance, $\phi(\cdot)$ denotes the layer mapping function between the student and teacher models, $U_{i,\phi(l)}^T$ and $U_{i,l}^S$ are the teacher and student span representations, respectively, and $w_{ij,l}^{\text{sp}} = w_{i,l}^{\text{sp}} \cdot w_{j,l}^{\text{sp}}$ prioritizes alignment between semantically salient event spans. By minimizing $\mathcal{L}_{EA}$, the student learns to preserve the within-layer relational structure among event spans, faithfully transferring the geometric relationships that encode the teacher’s event knowledge.

4.2 Training Objective

The complete training objective combines the token-level distillation loss with the event-aware loss:

$$\mathcal{L}_{\text{Total}} = (1 - \beta) \cdot \mathcal{L}_{\text{CE}} + \beta \cdot (\mathcal{L}_{\text{KD}} + \lambda_{EA} \cdot \mathcal{L}_{EA}) \quad (8)$$

where $\beta \in [0, 1]$ is the distillation ratio that balances the cross-entropy and distillation objectives, $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss defined in Equation 2, $\mathcal{L}_{\text{KD}}$ is the token-level distillation loss between teacher and student output distributions, and λ_{EA} controls the contribution of event-aware

alignment. We ablate the choice of $\mathcal{L}_{KD}$ using Reverse KL (RKL), Skew Forward KL (SFKL), and Concrete Score Distillation (CSD) in §5.

Conventional knowledge distillation relies solely on $\mathcal{L}_{KD}$, which operates at the output level by matching token-level probability distributions between the teacher and student. While this effectively transfers the teacher’s predictive behavior, it is agnostic to the internal representations that encode structural relationships among event components. In event extraction, the interdependencies among triggers, arguments, and event types carry essential semantic information that output-level alignment alone cannot capture. Our event-aware loss $\mathcal{L}_{EA}$ addresses this limitation by operating at the representation level, explicitly preserving the pairwise geometric relationships among event-aware spans across intermediate network layers. By combining both objectives, the student learns not only *what* the teacher predicts but also *how* the teacher internally organizes event-aware information, yielding richer and more structurally coherent event representations.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate on three widely used event extraction benchmarks: (1) **ACE05** (Dodington et al., 2004), which annotates 33 event types with both trigger and argument annotations across newswire and broadcast documents; (2) **GENEVA** (Parekh et al., 2023), a generative event argument extraction benchmark spanning diverse domains with both trigger and argument annotations; (3) **MAVEN** (Wang et al., 2020), a large-scale trigger detection dataset covering 168 event types from Wikipedia articles. For ACE05 and GENEVA, we report both trigger and argument extraction results; for MAVEN, which focuses on trigger-level evaluation, we report trigger metrics only.

Baselines. We compare against the following representative distillation baselines: (1) **SFT**: the student model fine-tuned on the same training data without any distillation signal; (2) **KD** (Hinton et al., 2015): standard knowledge distillation using forward KL divergence; (3) **RKL** (Wen et al., 2023): distillation using reverse KL divergence, which encourages mode-seeking behavior; (4) **SFKL** (Ko et al., 2024): Skew Forward KL divergence without span-level alignment, isolating the token-level component of our method; (5) **FDD**

(Gong et al., 2025): aligning both feature trajectories and their first-order layer-wise derivatives; (6) **DistiLLM** (Ko et al., 2024): a recent method that combines skewed divergences with adaptive off-policy training for LLM distillation; and (7) **CSD** (Kim et al., 2026): Concrete Score Distillation, which reformulates knowledge distillation through score matching over continuous relaxations of discrete tokens. We also include the **Teacher** model performance as an upper bound reference.

Implementation Details. We use Qwen3-4B-Instruct-2507 (Team, 2025) as the teacher model and Qwen3-0.6B as the student model. The teacher is first fine-tuned on each dataset using LoRA with rank 32, $\alpha = 64$, and dropout rate 0.1. For distillation, the student is trained with LoRA (rank 8, $\alpha = 16$, dropout 0.1) for 5 epochs. Both use a learning rate of 1×10^{-4} with cosine decay, weight decay of 1×10^{-2} , per-device batch size of 2, and gradient accumulation steps of 8 across 2 GPUs (effective batch size of 32). The maximum sequence length is set to 768 tokens. For the Skew Forward KL divergence, we instantiate the token-level distillation loss $\mathcal{L}_{KD}$ with the Skew Forward KL (SFKL) divergence (Ko et al., 2024), setting the skew parameter $\lambda = 0.1$. The distillation ratio is set to $\beta = 0.9$, so the total loss becomes $\mathcal{L}_{Total} = 0.1 \cdot \mathcal{L}_{CE} + 0.9 \cdot (\mathcal{L}_{KD} + \lambda_{EA} \cdot \mathcal{L}_{EA})$, where $\mathcal{L}_{KD}$ is instantiated as the SFKL divergence. For the Event-Aware Knowledge Distillation loss, we align intermediate Transformer layers between teacher and student. By default, we use two layers (teacher layers 33, 36 $\rightarrow$ student layers 25, 28); for ACE05, we use three layers (30, 33, 36 $\rightarrow$ 22, 25, 28), as the richer annotation schema benefits from deeper alignment. The span loss weight is set to $\lambda_{EA} = 2.0$. All experiments use DeepSpeed with bf16 mixed precision on two NVIDIA A100 40GB GPUs.

5.2 Event Extraction Results

Tables 1 and 2 present event extraction results. Table 1 reports event detection performance (trigger identification and type classification) across all three benchmarks. Because argument annotations are available only for ACE05 and GENEVA, Table 2 reports argument extraction performance (argument identification and role classification) on these two datasets.

Overall Performance. EventKD consistently achieves the highest F1 scores among all student

Table 1: Event detection results (Precision, Recall, F1) on three benchmarks. **Bold** indicates the best student result.

Method	ACE05			MAVEN			GENEVA		
	P	R	F1	P	R	F1	P	R	F1
Teacher	77.30	67.93	72.31	63.04	61.96	62.49	62.14	64.83	63.46
SFT	61.84	59.34	60.57	57.20	57.01	57.11	43.82	42.36	43.08
KD	68.02	59.09	63.24	61.72	58.06	59.84	54.43	47.81	50.90
RKL	67.23	60.61	63.75	65.05	57.16	60.85	51.85	46.75	49.17
SFKL	69.45	60.86	64.87	63.58	58.34	60.85	54.77	50.71	52.66
FDD	66.76	58.33	62.26	62.59	57.76	60.08	56.01	48.28	51.86
DistiLLM	66.86	57.58	61.87	63.97	58.31	61.01	54.55	50.45	52.42
CSD	70.62	60.10	64.94	65.21	57.11	60.89	56.86	49.76	53.07
EventKD	71.97	62.88	67.12	65.34	59.29	62.17	56.85	53.78	55.27

Table 2: Argument extraction results (Precision, Recall, F1) on ACE05 and GENEVA. **Bold** indicates the best student result.

Method	ACE05			GENEVA		
	P	R	F1	P	R	F1
Teacher	47.87	45.12	46.45	44.26	46.96	45.57
SFT	31.15	31.53	31.34	23.68	23.45	23.57
KD	34.76	33.97	34.36	34.66	30.88	32.66
RKL	34.77	33.62	34.19	30.95	29.85	30.39
SFKL	39.23	37.11	38.14	34.53	32.71	33.60
FDD	31.55	29.79	30.65	31.48	27.34	29.26
DistiLLM	36.35	34.32	35.30	34.61	32.90	33.73
CSD	38.74	35.37	36.98	36.16	33.07	34.54
EventKD	45.05	41.99	43.46	38.02	36.19	37.08

Table 3: Effect of layer mapping on ACE05. Teacher layers are mapped to corresponding student layers for span-level alignment.

Teacher	Student	Trigger F1	Argument F1
[]	[]	64.87	38.14
[36]	[28]	65.59	39.04
[33, 36]	[25, 28]	64.71	40.04
[27, 36]	[19, 28]	67.99	41.55
[28, 32, 36]	[20, 24, 28]	69.21	41.64
[30, 33, 36]	[22, 25, 28]	67.12	43.46
[27, 30, 33, 36]	[19, 22, 25, 28]	66.85	40.25

models across every dataset and metric. On ACE05, EventKD reaches 67.12% trigger F1 and 43.46% argument F1, outperforming the second-best method CSD (64.94% trigger F1) by 2.18 points and SFKL (38.14% argument F1) by 5.32 points. Compared to the SFT lower bound, EventKD improves trigger F1 by 6.55 points and argument F1 by 12.12 points, demonstrating the substantial benefit of distillation. Relative to the Teacher upper bound (72.31% / 46.45%), EventKD narrows the gap to 5.19 and 2.99 points on trigger and argument F1, respectively. On GENEVA, EventKD achieves 55.27% trigger F1 and 37.08% argument F1, surpassing all baselines. On MAVEN, EventKD reaches 62.17% trigger F1, approaching the teacher (62.49%) with only a 0.32-point gap.

Token-Level vs. Span-Level Distillation. To isolate the contribution of the Event-Aware loss $\mathcal{L}_{EA}$, we compare SFKL (token-level only) with SFKL + $\mathcal{L}_{EA}$ (our full objective). On ACE05, adding the span-level component improves trigger F1 from 64.87% to 67.12% (+2.25) and argument F1 from 38.14% to 43.46% (+5.32). The larger gain on arguments (+5.32 vs. +2.25) suggests

that span-level alignment is particularly beneficial for capturing argument structure, which involves richer relational dependencies than trigger detection alone. This pattern holds on GENEVA (+2.61 trigger F1, +3.48 argument F1), confirming that the structural relationships encoded by $\mathcal{L}_{EA}$ complement the token-level distributional signal across datasets.

5.3 Ablation Study

We conduct ablation experiments on ACE05 to analyze the contribution of each component in our framework. All experiments use the same teacher-student pair (Qwen3-4B / Qwen3-0.6B) and training configuration described in §5.1 unless otherwise noted.

Layer Mapping for Span Alignment. The event-aware loss $\mathcal{L}_{EA}$ aligns span representations extracted from intermediate Transformer layers of the teacher and student. Table 3 examines how the choice of aligned layers affects performance. Aligning no layers (row 1, $\lambda_{EA} = 0$) reduces to token-level distillation only. Using a single last layer provides modest gains, while mapping two evenly spaced layers (e.g., 27, 36 $\rightarrow$ 19, 28) yields stronger improvements, especially for trigger F1. The best overall balance on ACE05 is achieved by

Table 4: Effect of λ_{EA} on ACE05.

λ_{EA}	Trigger			Argument		
	P	R	F1	P	R	F1
0	69.45	60.86	64.87	39.23	37.11	38.14
0.5	73.28	64.39	68.55	42.31	38.85	40.51
1	72.38	62.88	67.30	44.46	41.99	43.19
2	71.97	62.88	67.12	45.05	41.99	43.46
3	72.22	62.37	66.94	40.95	39.02	39.96

Table 5: Training cost on ACE05 ($2 \times A100$ 40GB, batch size 16 per device).

Method	Time/step (s)	Avg alloc (GB)	Peak alloc (GB)
RKL	2.67	10.11	22.47
w/ $\mathcal{L}_{EA}$	2.77	10.96	23.32
SFKL	2.53	10.11	22.24
w/ $\mathcal{L}_{EA}$	2.65	10.96	22.62
CSD	2.39	10.11	18.75
w/ $\mathcal{L}_{EA}$	2.56	10.96	19.12

three layers (30, 33, 36 $\rightarrow$ 22, 25, 28), which we adopt for this dataset. Adding a fourth layer slightly degrades performance, suggesting that very early layers introduce noise rather than useful structural information.

Effect of λ_{EA} . Table 4 shows the effect of varying the event-aware loss weight λ_{EA} in the total objective. When $\lambda_{EA} = 0$, the model reduces to pure token-level distillation. Increasing λ_{EA} progressively improves both trigger and argument F1, with $\lambda_{EA} = 2$ achieving the best argument F1 (43.46) while maintaining competitive trigger F1 (67.12). Beyond this point ($\lambda_{EA} = 3$), argument performance drops noticeably, indicating that an overly strong span-level signal can interfere with token-level learning.

Choice of Token-Level Distillation Loss. Table 6 compares three token-level divergence functions – Reverse KL (RKL), Skew Forward KL (SFKL), and Concrete Score Distillation (CSD) – both with and without $\mathcal{L}_{EA}$. Without $\mathcal{L}_{EA}$, RKL yields the lowest F1, as its mode-seeking behavior causes the student to focus on a narrow subset of the teacher’s distribution, missing less frequent event types and roles. CSD achieves the highest trigger F1 (64.94) through smoother score-matching gradients, but lags on arguments (36.98 vs. 38.14 for SFKL), suggesting that continuous relaxation is less effective for discrete argument structures. SFKL strikes the best balance by interpolating a mixture distribution that stabilizes gradients while preserving mode-covering behavior. Adding $\mathcal{L}_{EA}$

Table 6: Comparison of token-level KD losses on ACE05, with and without the event-aware loss $\mathcal{L}_{EA}$.

Method	Trigger			Argument		
	P	R	F1	P	R	F1
Teacher	77.30	67.93	72.31	47.87	45.12	46.45
SFT	61.84	59.34	60.57	31.15	31.53	31.34
RKL	67.23	60.61	63.75	34.77	33.62	34.19
w/ $\mathcal{L}_{EA}$	71.14	62.88	66.76	42.48	38.85	40.58
SFKL	69.45	60.86	64.87	39.23	37.11	38.14
w/ $\mathcal{L}_{EA}$	71.97	62.88	67.12	45.05	41.99	43.46
CSD	70.62	60.10	64.94	38.74	35.37	36.98
w/ $\mathcal{L}_{EA}$	72.19	64.90	68.35	40.91	39.20	40.04

improves all three divergences, with the largest gains on RKL (+6.39 argument F1), indicating that weaker token-level objectives benefit more from span-level structural guidance. We adopt SFKL as the default because it achieves the best argument F1 (43.46) with $\mathcal{L}_{EA}$, and argument extraction is the harder subtask.

Computational Overhead. Table 5 shows the training cost of adding $\mathcal{L}_{EA}$ on ACE05 using 2 A100 40GB GPUs (batch size 16). The overhead is modest: step time increases by $<5\%$ (2.53 $\rightarrow$ 2.65s for SFKL) and memory by 0.85GB, with no change in peak allocation tier. Overall, $\mathcal{L}_{EA}$ is computationally efficient and easy to integrate.

6 Conclusion

We presented EventKD, a two-level knowledge distillation framework for event extraction that addresses key limitations of standard token-level objectives. At the token level, a general distributional divergence loss $\mathcal{L}_{KD}$, instantiated in our experiments with the Skew Forward KL (SFKL) divergence, aligns teacher and student output distributions with improved stability over standard forward and reverse KL. At the span level, an Event-Aware (EA) Knowledge Distillation loss preserves pairwise geometric relationships among event-aware spans, including triggers, arguments, and event types, across multiple Transformer layers. By combining both objectives, EventKD enables a compact student model to capture not only the teacher’s output behavior but also the internal structural knowledge critical for event understanding. Experiments on three benchmarks (ACE05, GENEVA and MAVEN) demonstrate consistent improvements over existing distillation methods on both trigger identification and argument extraction.

7 Limitations

Our framework has several limitations that suggest directions for future work. First, the span extraction step relies on the structured output format of the teacher model to identify triggers, arguments, and event types; if the teacher produces malformed or incomplete outputs, the resulting span boundaries may be noisy, potentially degrading the quality of the EA loss signal. Second, our experiments focus on English-language benchmarks, and the extent to which span-level alignment transfers to morphologically rich or low-resource languages, where event structure may differ substantially, remains an open question. Third, while we evaluate on three standard EE datasets, the present study does not explore interactions between our distillation objectives and complementary compression techniques such as quantization or pruning, which may offer additional efficiency gains. Finally, computing pairwise distances among all extracted spans introduces quadratic overhead in the number of spans; for documents with many events, efficient approximations or sparse attention mechanisms may be necessary to maintain scalability.

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A Appendix

A.1 Token Importance Computation

This section provides a detailed derivation of the token importance weights $w_{t,l}$ used in the span representation (Eq. 4). Rather than treating all tokens within a span equally, we assign each token a scalar importance score that reflects how much contextual relevance it carries, as determined by the collective attention it receives from other positions in the sequence. This mechanism ensures that semantically salient tokens, such as trigger words and argument heads, contribute more prominently to the span-level representation.

Hidden State Normalization. Given the hidden state $H_{t,l} \in \mathbb{R}^d$ of token t at Transformer layer l , we first apply variance-based normalization to obtain a standardized representation:

$$\hat{H}_{t,l} = \frac{H_{t,l}}{\sigma(H_{t,l})}, \quad (9)$$

where $\sigma(H_{t,l})$ denotes the standard deviation computed across the d dimensions of $H_{t,l}$. This normalization step removes magnitude differences across tokens and layers, allowing the subsequent attention computation to focus purely on directional similarity between token representations.

Pairwise Attention Scoring. We compute a pairwise relevance score between every pair of tokens (s, t) in the sequence using scaled dot-product attention on the normalized hidden states. To prevent trivial self-reinforcement and to handle variable-length sequences, we apply a masking function that excludes self-loops and padding positions:

$$S_{s \rightarrow t,l} = \frac{\hat{H}_{s,l}^\top \hat{H}_{t,l}}{\sqrt{d}} + M_{s,t}, \quad (10)$$

where the mask $M_{s,t}$ is defined as:

$$M_{s,t} = \begin{cases} -\infty & \text{if } s = t \text{ or } t \in \mathcal{P}, \\ 0 & \text{otherwise,} \end{cases} \quad (11)$$

and $\mathcal{P}$ is the set of padding token indices. Setting $M_{s,s} = -\infty$ eliminates self-attention, ensuring that a token’s importance is determined solely by how other tokens attend to it. Masking out padding positions prevents them from influencing the importance scores.

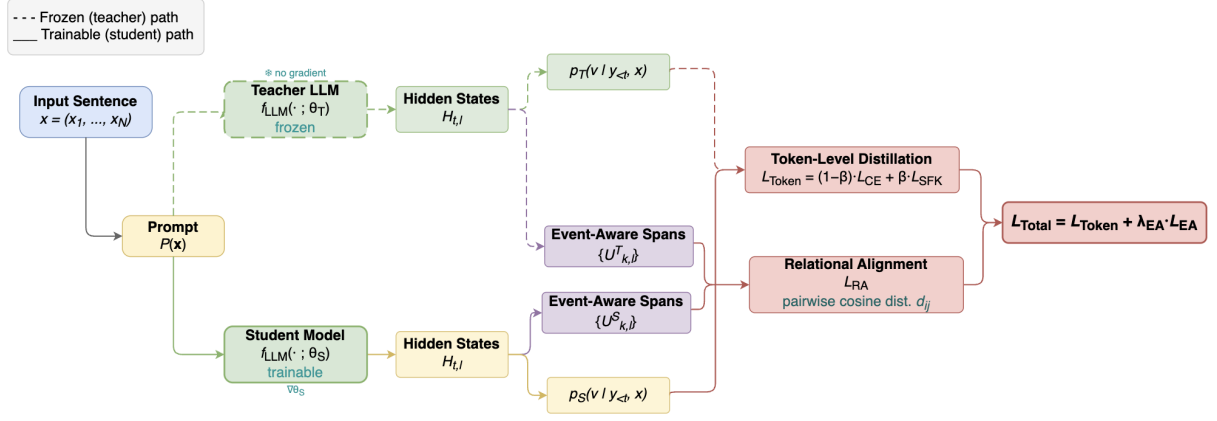


Figure 3: Overview of our knowledge distillation framework for LLM-based event extraction. The input sentence is processed through a prompt template and fed to both the frozen Teacher LLM and the trainable Student model. Token-level distillation ($\mathcal{L}_{\text{KD}}$) aligns output distributions using Skew Forward KL divergence. Structural distillation ($\mathcal{L}_{\text{EA}}$) extracts event-aware spans from hidden states and aligns their pairwise relational geometry between teacher and student.

Attention Weight Computation. The raw scores are converted into proper probability distributions via a row-wise softmax over valid (non-masked) positions:

$$\alpha_{s \rightarrow t, l} = \frac{\exp(S_{s \rightarrow t, l})}{\sum_{u=1}^N \exp(S_{s \rightarrow u, l})}, \quad (12)$$

where N is the total number of non-padding tokens. The coefficient $\alpha_{s \rightarrow t, l}$ quantifies the degree to which token s attends to token t at layer l .

Importance Aggregation. Finally, the importance weight of token t at layer l is obtained by averaging the attention it receives from all other tokens in the sequence:

$$w_{t, l} = \frac{1}{N} \sum_{s=1}^N \alpha_{s \rightarrow t, l}. \quad (13)$$

Intuitively, $w_{t, l}$ measures the mean inbound attention directed toward token t : tokens that are consistently attended to by many other positions receive higher importance scores. This attention-based aggregation naturally assigns greater weight to tokens that serve as central anchors in the contextual representation, such as event triggers and salient argument tokens, making it well-suited for event-aware span pooling in Eq. 4.